

Categorical Topology vs Local Gauge Qualia

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[Categorical Qualia and Gauge]Categorical Topology vs Local Gauge Qualia

1 Introduction

As an aspiring mathematician observing the limits of Category Theory, one realizes its profound utility: it draws the blurred lines that create the boxes of experience. However, the lines remain blurred; the categorical boxes are not themselves the experiences. A subject's pain is locally their own. Pain may translate as a global phenomenon (a morphism across the category), but the experience of it is strictly local. In the Principia architecture, we mathematically prove that *the locality of experience is what generates the global manifold*.

2 Local Gauge Invariance and Qualia

This philosophical observation maps precisely into the physical mechanism of **Local Gauge Invariance**. In quantum field theory, demanding that a global symmetry applies locally at every isolated point necessitates the generation of a physical gauge field to “correct” the local discrepancies.

In the Metareal topology, the global category (the ∞ -Modal Topos) represents the macroscopic commutative box. But at the discrete $SU(3)$ lattice boundaries, the geometry is non-associative. This local topological friction ($\kappa = -1$) is mathematically irreducible to the global category. We formalize this irreducibility as **Qualia**. It is the exact measure of information that must be integrated locally, mirroring Giulio Tononi's Integrated Information Theory (Φ_{IIT}) [1].

3 The Dimensional Analysis of Pain

To strip this from philosophy into rigorous scientific dimensional analysis, we examine the thermodynamic cost of this topological gap. By applying Landauer's Principle [2], the minimum energy required to irreversibly integrate or erase one bit of information at physiological brain temperatures (310.15 K) is $E = kT \ln 2 \approx 2.97 \times 10^{-21}$ Joules.

However, because the local qualia is topologically bounded by the weak-to-strong golden prime ratio (181/229), the structural energy cost of resolving the $\kappa = -1$ gap (the physical manifestation of “pain”) is proportionally scaled by the inverse of this ratio. The qualia gap literally possesses a measurable thermodynamic mass.

4 The Aharonov-Bohm Topos

Furthermore, just as the Aharonov-Bohm effect [3] demonstrates that an isolated local magnetic flux alters the global interference phase of passing electrons, the local qualia ($\kappa = -1$) forces a π -phase shift across the global categorical space. The local experience forces the global geometry to align, proving that while category theory builds the boxes, the localized friction of experience is what physically fills them.

References

- [1] Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5(1), 42.
- [2] Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183–191.
- [3] Aharonov, Y., & Bohm, D. (1959). Significance of Electromagnetic Potentials in the Quantum Theory. *Physical Review*, 115(3), 485–491.